ReDel as a POMDP

Learning to Delegate

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$$\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathbb{T}, \mathbb{T}_0, R \rangle$$

- $s_t \in \mathcal{S}$: the current agents/sub-agents

- $\mathscr{A} = \{\text{Delegate(task)}, \text{Solve(text)}\} \cup \mathscr{A}_{\text{tools}}$: the actions each agent can take

- $o_t \in \mathcal{O}$: result of a tool or a sub-agent's solution

- $s_{t+1} \sim \mathbb{T}(s_{t+1} \mid s_t, a_t)$: the transition dynamics of the system: add agents on Delegate(), remove agent or end on Solve()

- $\mathbb{T}_0$: the distribution of starting states (user prompt + 1 root agent)

- $R(s_t)$: the reward function

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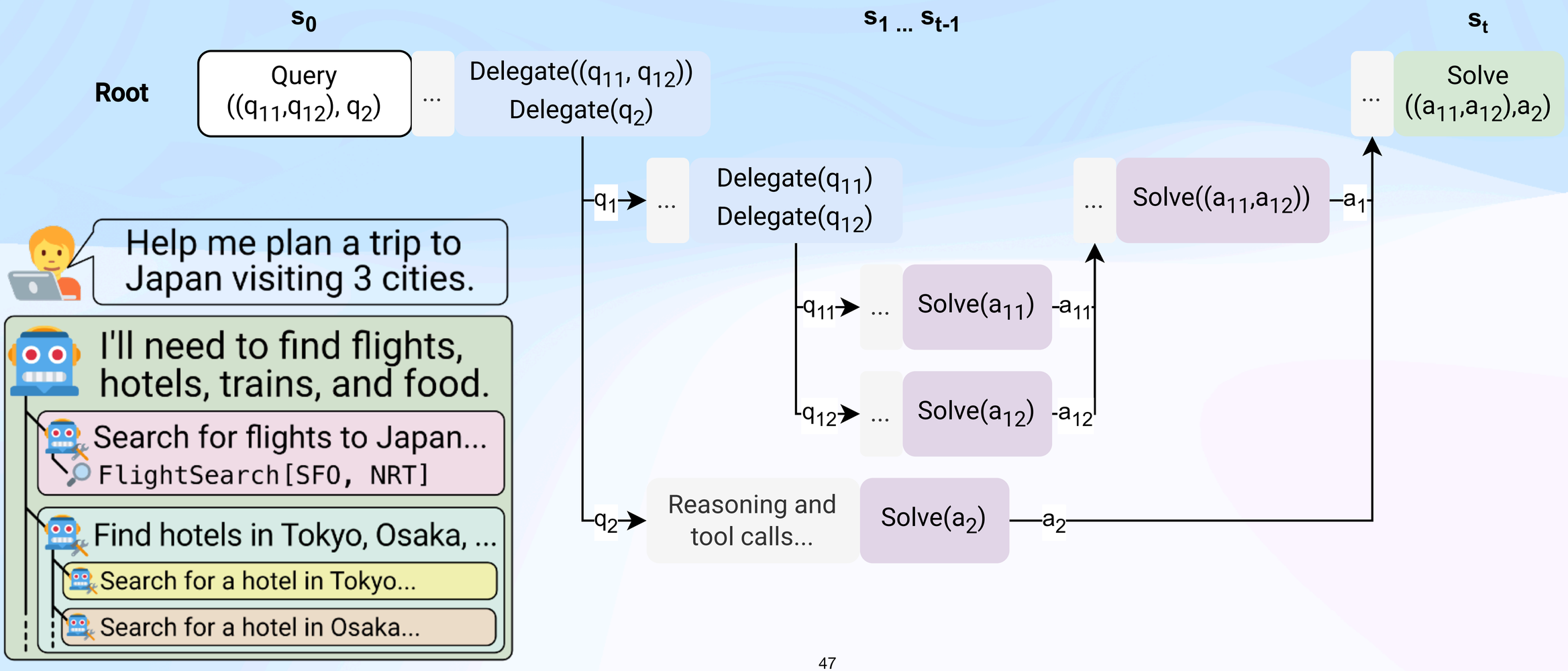
RL Example

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